Prioritizing potential marine aquaculture

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This analysis examines which Exclusive Economic Zones (EEZ) on the United States' West Coast are best suited for marine aquaculture of oysters and other species. Location suitability is determined based on the species' optimal range of sea surface temperature (SST) and depth.

Sea surface temperatures used in this analysis are obtained from NOAA's 5km Daily Global Satellite Sea Surface Temperature Anomaly v3.1. Measurements are averaged over four years between 2008 and 2012. Bathymetry measurements are taken from the General Bathymetric Chart of the Oceans (GEBCO). Boundaries of Exclusive Economic Zones are designated based on Marineregions.org.

Load and Process Data

```
# Load Packages
library(terra) # raster handling
library(tidyverse)
library(tmap)
library(spData) # spatial data
library(spDataLarge) # spatial data
library(stars)
library(stringr)
```

```
# Load data
depth <- rast("/home/oknapp/eds223/EDS223-HW3/data/depth.tif")
sst_2008                <-                rast("/home/oknapp/eds223/EDS223-HW3/
data/average_annual_sst_2008.tif")
sst_2009                <-                rast("/home/oknapp/eds223/EDS223-HW3/
data/average_annual_sst_2009.tif")
sst_2010                <-                rast("/home/oknapp/eds223/EDS223-HW3/
data/average_annual_sst_2010.tif")
sst_2011                <-                rast("/home/oknapp/eds223/EDS223-HW3/
data/average_annual_sst_2011.tif")
sst_2012                <-                rast("/home/oknapp/eds223/EDS223-HW3/
data/average_annual_sst_2012.tif")
wc_regions           <-           st_read("/home/oknapp/eds223/EDS223-HW3/data/
wc_regions_clean.shp")
```

```
Reading layer `wc_regions_clean' from data source
  `/home/oknapp/eds223/EDS223-HW3/data/wc_regions_clean.shp'
  using driver `ESRI Shapefile'
Simple feature collection with 5 features and 5 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -129.1635 ymin: 30.542 xmax: -117.097 ymax: 49.00031
Geodetic CRS:  WGS 84
```

```
#Check that SST file crs match
if (same.crs(sst_2008, sst_2009)&
    same.crs(sst_2008, sst_2010)&
    same.crs(sst_2008, sst_2011)&
    same.crs(sst_2008, sst_2012)){
    return(invisible(NULL))
} else{
    warning("CRS do not match. Transform before joining")}
```

```
# Combine sst files, take 4-year mean, and convert to celsius
sst_mean_c <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012) |>
    mean(na.rm = TRUE) - 273.15

#Adjust the crs of the depth raster to match the crs of the sst raster
if (crs(sst_mean_c) == crs(depth)){
    return(invisible(NULL))
}else{
    crs(depth) <- crs (sst_mean_c)}

#Resample the depth data to match the resolution of the SST and crop to match
the extent of SST
depth_cropped <- resample(depth, sst_mean_c, method = "bilinear") |>
                    crop(sst_mean_c)

#Confirm crs, resolutions and extents match
if(same.crs(sst_mean_c, depth_cropped)){
    return(invisible(NULL))
} else {
    warning("depth and sst raster crs do not match")}
if(ext(depth_cropped) == ext(sst_mean_c)){
    return(invisible(NULL))
} else {
    warning("depth and sst raster extents do not match")}
```

```
if(mean(res(depth_cropped) == res(sst_mean_c)) == 1) {
    return(invisible(NULL))
} else {
    warning("depth and sst raster resolutions do not match")}
```

Find Suitable Locations

The following function can be used to identify suitable locations for marine farming of a given species and their temperature and depth range. It relies on the previously-processed data to provide a map of exclusive economic zones of the west coast of the United States, shaded by total area suitable for farming the given species. Precise locations within zones with appropriate habitats for farming are indicated in red.

Two examples are given below; oysters and red abalone.

```
# Create a function to map suitable locations for farming given a species,
suitable_farming_locations <- function(species, temp_min, temp_max, depth_min,
depth_max) {
    if (depth_max > 0) {
        warning ("Depths expected as negative values or 0")}

    # Categorize suitable temperatures
    sst_rcl <- matrix(c(
       -Inf, temp_min, 0,
        temp_min, temp_max, 1,
        temp_max, Inf, 0
    ), ncol = 3, byrow = TRUE)
    sst_zones <- classify(sst_mean_c, rcl = sst_rcl)
    # Categorize suitable depths
    depth_rcl <- matrix(c(
        -Inf, depth_max, 0,
        depth_max, depth_min, 1,
        depth_min, Inf, 0
    ), ncol =3, byrow = TRUE)
    depth_zones <- classify(depth_cropped, rcl = depth_rcl)

    #Create a mask to identify suitable farming areas
     suitable <- c(sst_zones, depth_zones) |> lapp(fun = function(sst, depth)
sst*depth)

    suitable_zones_mask <- cellSize(suitable, unit = 'km', mask = TRUE) |>
        mask(suitable, maskvalues = 0)
```

```
    # Make the crs of EEZ locations match crs of suitable locations
    if (crs(wc_regions) != crs(suitable)){
         sf::st_crs(wc_regions) <- terra::crs(suitable)
    }

    # Convert EEZ location data into a raster
    eez_suitable <- rasterize(wc_regions, suitable, field = 'rgn_id')

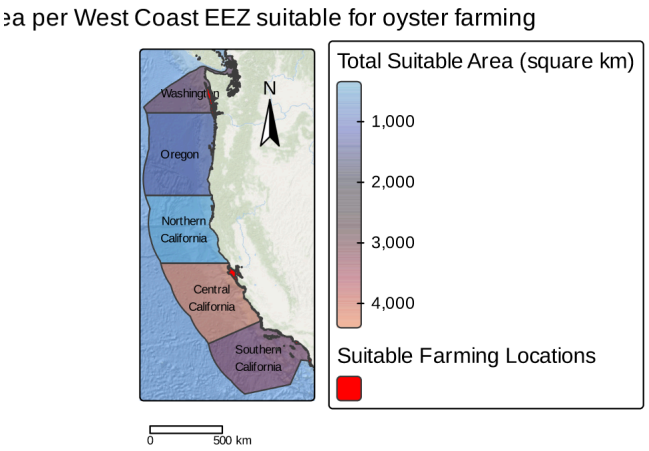
    # Calculate total area covered by suitable locations
    eez_suitable_areas <- zonal(suitable_zones_mask, eez_suitable, fun = "sum",
na.rm = TRUE) |>
        left_join(x = wc_regions, by = 'rgn_id') |>
          mutate(rgn = str_replace_all(rgn, " ", "\n") #Format EEZ names for
mapping labels
    )

    # Create polygons to map suitable farming locations
    suitable_locations <- as.polygons(suitable_zones_mask, dissolve = TRUE)

    # Map EEZs by total suitable area
    tmap_options(show.messages = FALSE,
                 show.warnings = FALSE)
    tm_basemap("Esri.OceanBasemap") +
    tm_options(component.autoscale= FALSE) +
    tm_shape(eez_suitable_areas) +
        tm_polygons(fill = 'area',
                            fill.legend = tm_legend(title = "Total Suitable Area
(square km)"),
                    fill.scale = tm_scale_continuous(values = 'seaborn.icefire'),
                      fill_alpha = 0.5) +
        tm_text(text = 'rgn',
                size = 0.5,
                col = 'black') +
    tm_shape(suitable_locations) +
        tm_polygons(
            fill = 'red',
            fill.legend = tm_legend(title = "Suitable Farming Locations")) +
    tm_compass(position = tm_pos_in("right", "top")) +
    tm_scalebar(position = tm_pos_out("center", "bottom")) +
    tm_legend(postion = tm_pos_out('right', 'center')) +
    tm_add_legend(fill = 'red',
                  title = "Suitable Farming Locations",
                  type = 'polygons',
```

```
                  position = tm_pos_out('right', 'center')) +
      tm_title(paste("Total area per West Coast EEZ suitable for", species,
"farming"),
               size = 1,
               position = tm_pos_out("center", "top"))
    }
```

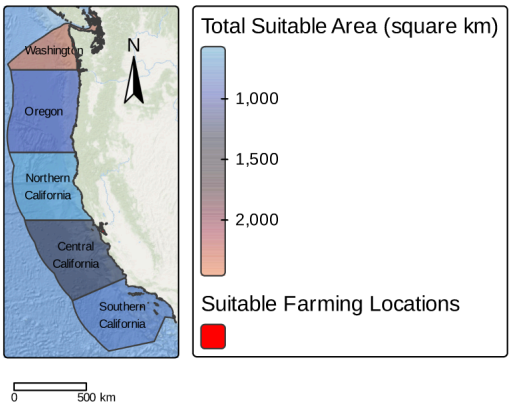
```
# Identify suitable farming locations for oysters
suitable_farming_locations('oyster', temp_min = 11, temp_max = 30, depth_min =
0, depth_max = -70)
```



Oysters have an optimal temperature range of 11 to 30 degrees Celsius, and an optimal depth range of up to 70 meters below sea level. Considering both of these factors, Central California has the greatest total area suitable for oyster farming, followed by Southern California and Washington. Northern California is the least suited. The majority of the suitable area in Central California lies in the San Francsisco and Monterey Bay areas, though appropriate locations can be found throughout much of the coastline.

```
# Identify suitable farming locations for red abalone
suitable_farming_locations('red abalone', temp_min = 8, temp_max = 18, depth_min
= 0, depth_max = -24)
```

ɔer West Coast EEZ suitable for red abalone farming



Red abalone have an optimal temperature range of 8 to 18 degrees Celsius, and an optimal depth range of up to 24 meters below sea level. Considering both of these factors, Washington is the best suited for abalone farming, followed by Central California. Northern California is the least suitable. The majority of Washington's coastline is suitable for red abalone, particularly the coastline of the bays off of the Rosario Strait and Strait of Georgia. In Central California, suitable locations are restricted primarily to the San Francisco Bay, though narrow areas are suitable along much of the west coast.

References

- Coral Reef Watch. (2018, August 1). Daily Global 5km Satellite Sea Surface Temperature Anomaly (Version 3.1). NOAA Coral Reef Watch Daily 5km satellite coral bleaching heat stress SST anomaly product (version 3.1). https://coralreefwatch.noaa.gov/product/5km/index_5km_ssta.php
- Flanders Marine Institute. (n.d.). EEZ boundaries. Marine regions. https://www.marineregions.org/eez.php
- GEBCO Compilation Group (2022) GEBCO_2022 Grid (doi:10.5285/e0f0bb80-ab44-2739-e053-6c86abc0289c).
- Palomares, M.L.D. and D. Pauly. Editors. 2026. SeaLifeBase. World Wide Web electronic publication. www.sealifebase.org, version (06/2026).